## Daily requirements

The ICMR recommends 1gm of protein/ Kg of body weight for adults.

The amount of protein should be increased for children, pregnant and lactating mothers by 1.5- 2 g/kg body weight.



## Sources of Protein

**Animal sources:** eggs, milk and milk products, fish, meat.

**Plant sources:** pulses, cereals, dry fruits, beans nuts etc.

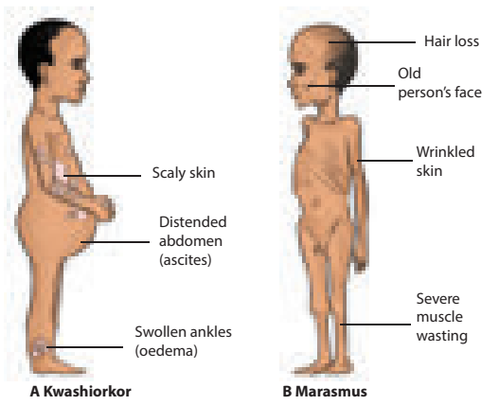
## Functions of Proteins

- Protein helps in synthesis of enzymes, immunoglobulin, plasma proteins and hormones in the body
- Protein helps in growth and repair of body tissues
- Proteins are secondary sources of energy during deficiency of carbohydrates and fats. (Provides 4 Kcals/gm)
- Proteins help in forming haemoglobin
- Proteins help in antibody formation

**Digestion and absorption:** The hydrolysis of proteins in the gastro intestinal tract is accomplished by proteases secreted in gastric juice and pancreatic juice and also by proteases present in the intestinal mucosa.




Severe protein deficiency along with food energy causes Protein Calorie Malnutrition (PCM). The serious forms of PCM are Kwashiorkor and Marasmus.




**A Kwashiorkor** **B Marasmus**

## III. Fats

Fat is deposited as adipose tissue in the body and perform essential functions in the body. Fats are composed of fatty acids and contain oxygen, carbon and hydrogen.

Classification: Fats are classified in to two types: saturated fat and unsaturated fat.

a) **Saturated Fat:** These have full number of hydrogen atoms. These are from animal sources. **Example:** butter, Ghee and vanaspati oil.

b) **Unsaturated Fat:** These contain one, two or more double bonds of fatty acids. These are extracted from vegetable sources. **Example:** Groundnut oil, soyabean oil, sunflower oil.




Saturated fats changes in to solid from at low temperature and melts at high temperature. Unsaturated fats never solidify at any temperature.



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